Image Generation Models

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected image-evidence record.

Explain origin, inspect the poster, preserve the qualification and design a visual check.

Keep one habit: Retain supplied arrays, criterion evidence, production history and unmeasured claims separately.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Prompt Evaluation

Iterative Prompting

Sampling and Variability

W1 · Explain generation and provenance

> Distinguish a constructed picture from an observed event.

Your evidence record

- Poster audit: required exactly3readable circles A,B,C;constructed candidate has4circles A,B,illegible,D and a draft photograph caption. Grid process is a separate supplied2×2before/after illustration; it is not a trained diffusion run and supplies no clean target or quality score. Provenance documents production history, not the truth of every depicted fact.

- Reference for this exercise: open this week's student lesson, "Image Generation Models", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Audit the complete poster candidate

> Check count, labels and the draft photograph claim separately.

Your evidence record

- Poster audit: required exactly3readable circles A,B,C;constructed candidate has4circles A,B,illegible,D and a draft photograph caption. Grid process is a separate supplied2×2before/after illustration; it is not a trained diffusion run and supplies no clean target or quality score. Provenance documents production history, not the truth of every depicted fact.

- A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

  Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Preserve the provenance qualification

> A production record cannot establish every factual or permission claim.

Your evidence record

- Poster audit: required exactly3readable circles A,B,C;constructed candidate has4circles A,B,illegible,D and a draft photograph caption. Grid process is a separate supplied2×2before/after illustration; it is not a trained diffusion run and supplies no clean target or quality score. Provenance documents production history, not the truth of every depicted fact.

- A report omits this qualification: **Provenance** tracks production history; it does not prove the content is accurate or establish every legal permission.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Design a revealing visual check

> State the expected criterion result before changing a specification or drawing.

Your evidence record

- Poster audit: required exactly3readable circles A,B,C;constructed candidate has4circles A,B,illegible,D and a draft photograph caption. Grid process is a separate supplied2×2before/after illustration; it is not a trained diffusion run and supplies no clean target or quality score. Provenance documents production history, not the truth of every depicted fact.

- Reference for this exercise: open this week’s student lesson, “Image Generation Models”, at “See how it works” and “Follow a worked example”. Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Audit a new diagram independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Audit the supplied description of the independent diagram. No actual image or camera record is provided, so do not invent geometry or an observed experiment. Treat arrow count, label meaning and the authenticity caption as separate requirements.

A generated diagram requires two arrows and labels **input**/**output**. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental **observation**. Audit.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Treat generated diagrams as illustrations; pair factual claims with approved sources.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Image generation and **provenance**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Image generation and **provenance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

An image generator produces a convincing picture of a school greenhouse on Mars. A student calls it photographic **evidence**. Explain what the image can and cannot establish.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A generator creates a realistic fictional bicycle beside a tree.

How does this differ from event photography?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

An image model draws a plant with seven roots after a request for exactly four. What does this show, and where should an exact count diagram be made?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 10 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

An image generator produces a convincing picture of a school greenhouse on Mars. A student calls it photographic **evidence**. Explain what the image can and cannot establish.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A generated diagram requires two arrows and labels **input/output**. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental **observation**. Audit.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

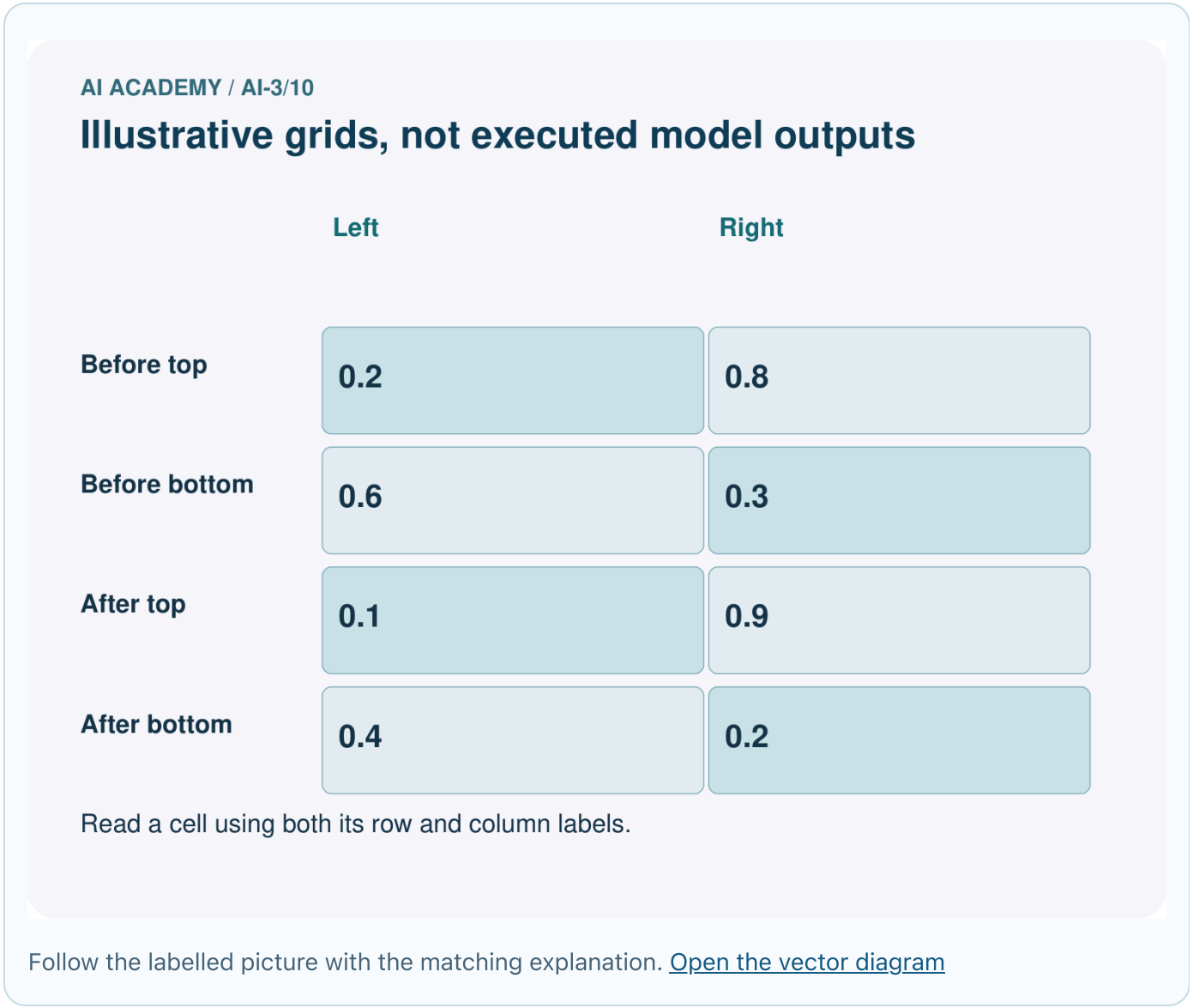
> Keep the explanation and the support you used.

Use this when helpful

A fictional lesson poster illustrates one family of image-generation approaches.

Toy 2×2 noisy numerical grid: [[0.2,0.8],[0.6,0.3]]. A specified illustrative **denoising** step gives [[0.1,0.9],[0.4,0.2]].

These supplied arrays illustrate steps only; they are not **output**s from a real trained diffusion model.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/10

Image Generation Models

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Noisy representation

Four toy numerical cells form the starting grid.



2. Condition on request

A text request guides the imagined learned denoising process.



3. Illustrative update

Use the supplied second grid to show one change in representation.



4. Check provenance

Caption it as a generated-process illustration, not a photographed experiment.

CHANGE ONE THING

Relabel the illustration as a photograph of an actual laboratory measurement.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

What may these two grids explain, and what can they not establish about a real generator?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Use Study Buddy generated diagrams only with clear **provenance** and checked educational labels.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use Study Buddy generated diagrams only with clear **provenance** and checked educational labels.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional lesson poster illustrates one family of image-generation approaches.
- Which labelled step matches this detail? Caption it as a generated-**process** illustration, not a photographed experiment.
- 1. Noisy representation
- 2. Condition on request
- 3. Check **provenance**
- 4. Illustrative update
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Check **provenance** Caption it as a generated-**process** illustration, not a photographed experiment.

Diffusion-style generators learn to remove **noise** in stages, often guided by text, but a real implementation can operate in a **latent representation** rather than direct pixels.

The toy values let us inspect an update without pretending to simulate that trained **process**. An attractive final image still needs checks for labels, requested relations and factual claims.

Changed-case question

- Changed-case practice: Relabel the illustration as a photograph of an actual laboratory measurement. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. That caption is unsupported.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: That caption is unsupported. Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

Repair and save your work

Review Prompt Evaluation (ai-3/9). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Why is B's selection defensible here without calling it universally superior?.

Return to this week and explain what you changed.

Review Iterative Prompting (ai-3/7). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: What **evidence** supports improvement here, and what remains unproven?.

Return to this week and explain what you changed.

Review Sampling and Variability (ai-3/4). Goal: Sample from a distribution using explicit cumulative intervals. Short practice: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

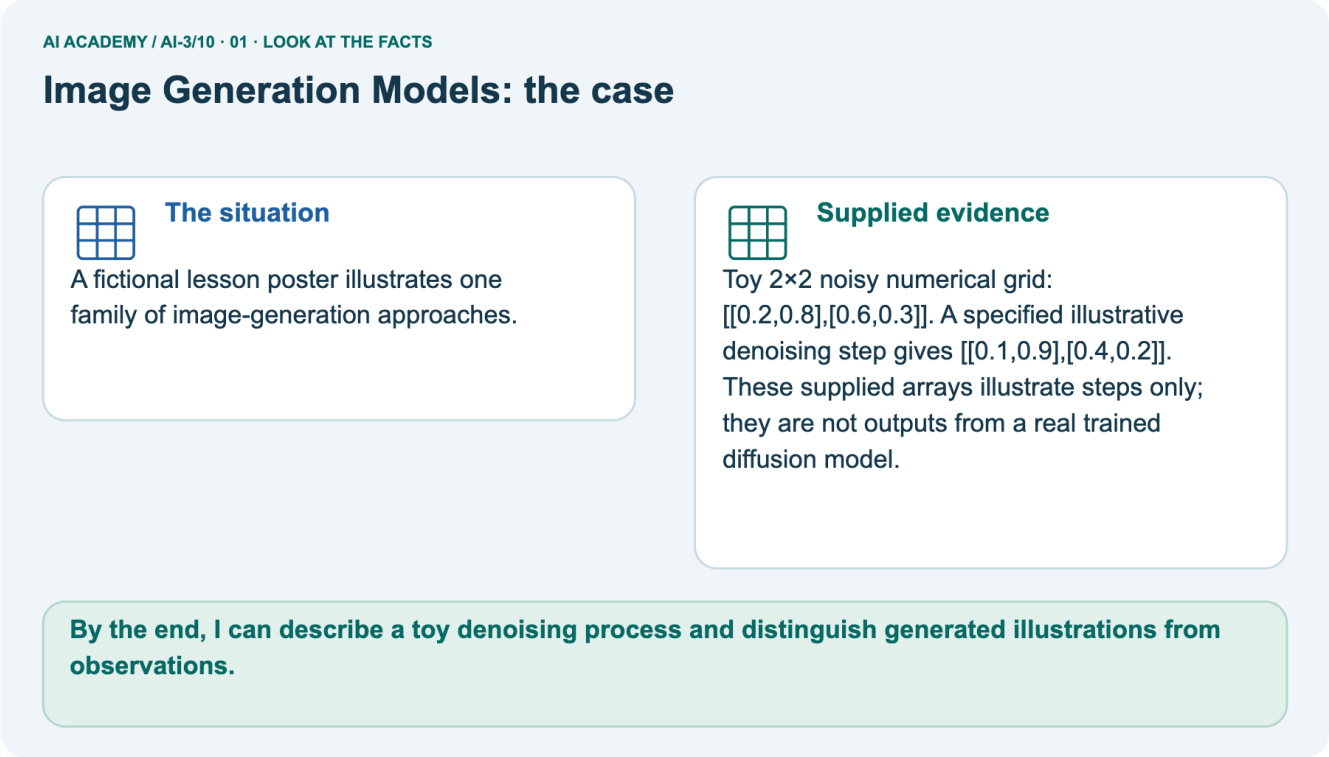
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

Choose the image that helps rather than repeating all six.

01. Image Generation Models: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

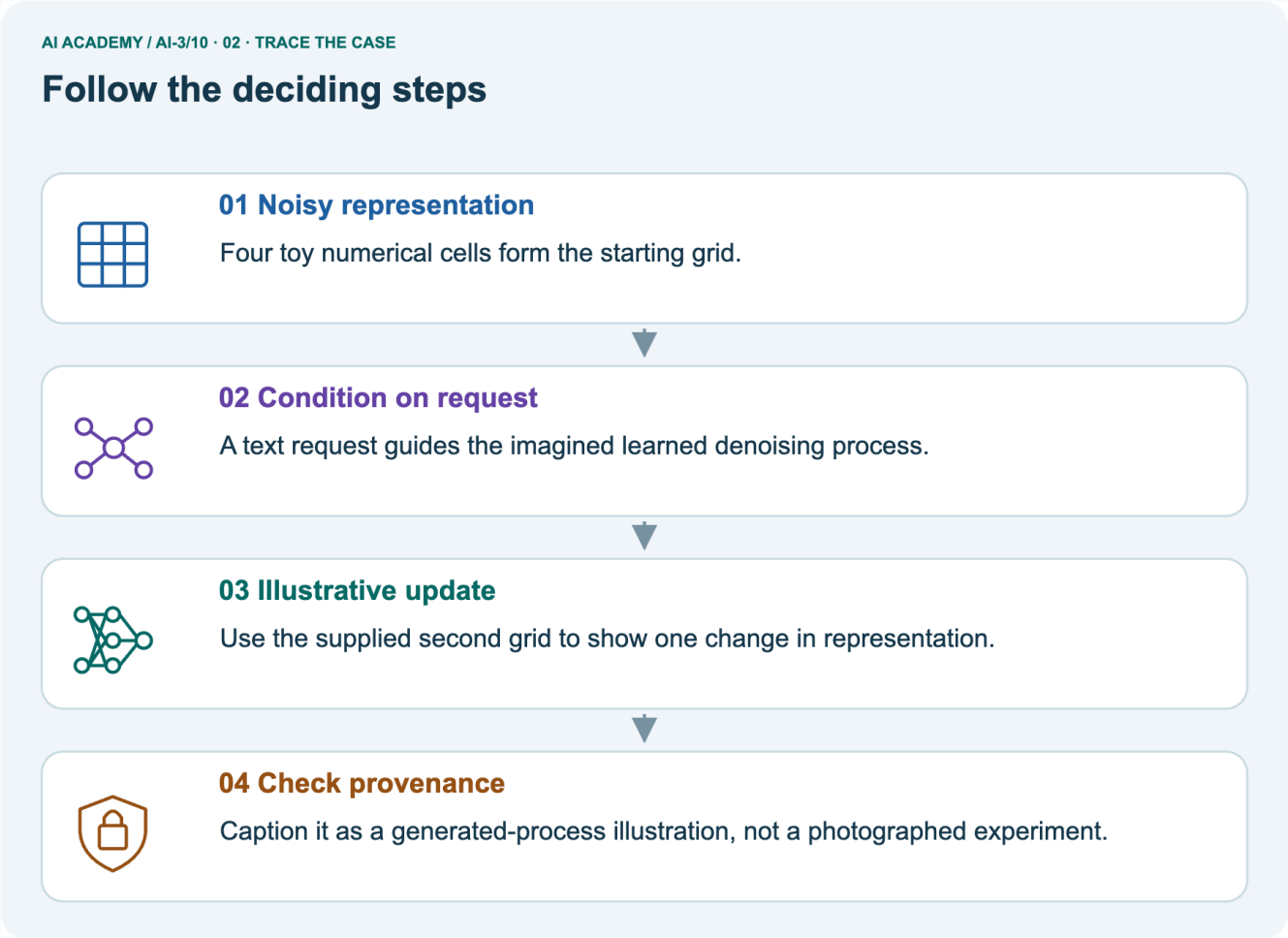
A fictional lesson poster illustrates one family of image-generation approaches. Supplied facts: Toy 2×2 noisy numerical grid: [[0.2,0.8],[0.6,0.3]]. A specified illustrative **denoising** step gives [[0.1,0.9],[0.4,0.2]].

These supplied arrays illustrate steps only; they are not **output**s from a real trained diffusion model.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



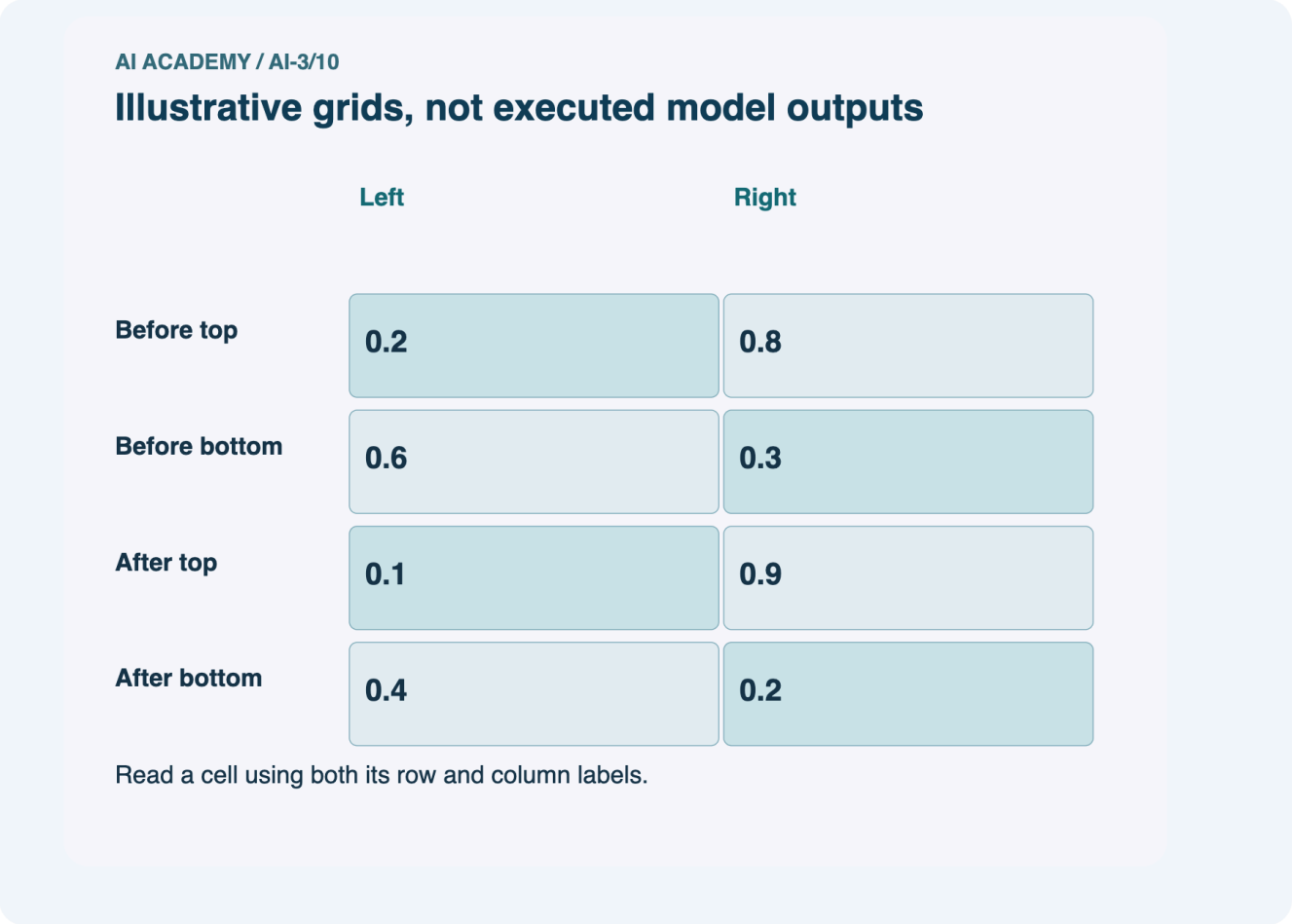
Follow the labelled picture with the matching explanation. Open the vector diagram

Noisy representation: Four toy numerical cells form the starting grid. → Condition on request: A text request guides the imagined learned **denoising process**. → Illustrative update: Use the supplied second grid to show one change in representation. → Check **provenance**: Caption it as a generated-**process** illustration, not a photographed experiment.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Illustrative grids, not executed model outputs



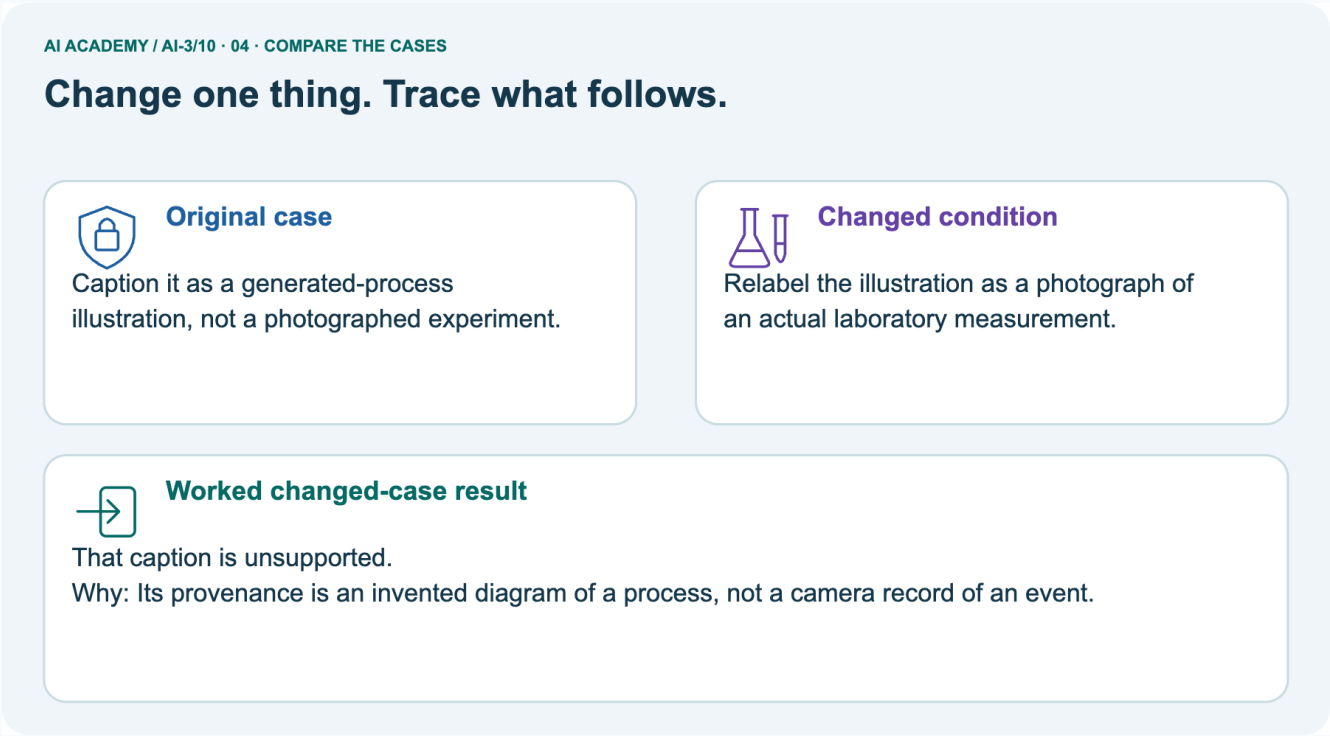
Follow the labelled picture with the matching explanation. Open the vector diagram

Illustrative grids, not executed model **output**s. Before top: Left = 0.2, Right = 0.8. Before bottom: Left = 0.6, Right = 0.3. After top: Left = 0.1, Right = 0.9. After bottom: Left = 0.4, Right = 0.2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Caption it as a generated-**process** illustration, not a photographed experiment. Changed condition: Relabel the illustration as a photograph of an actual laboratory measurement.

Expected result: That caption is unsupported. Reason: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit

AI ACADEMY / AI-3/10 · 05 · EXPLAIN WITH CARE

Connect the explanation to its limit

What the case explains

Diffusion-style generators learn to remove noise in stages, often guided by text, but a real implementation can operate in a latent representation rather than direct pixels. The toy values let us inspect an update without pretending to simulate that trained process. An attractive final image still needs checks for labels, requested relations and factual claims.

What it does not establish

Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

Follow the labelled picture with the matching explanation. Open the vector diagram

Diffusion-style generators learn to remove **noise** in stages, often guided by text, but a real implementation can operate in a **latent representation** rather than direct pixels.

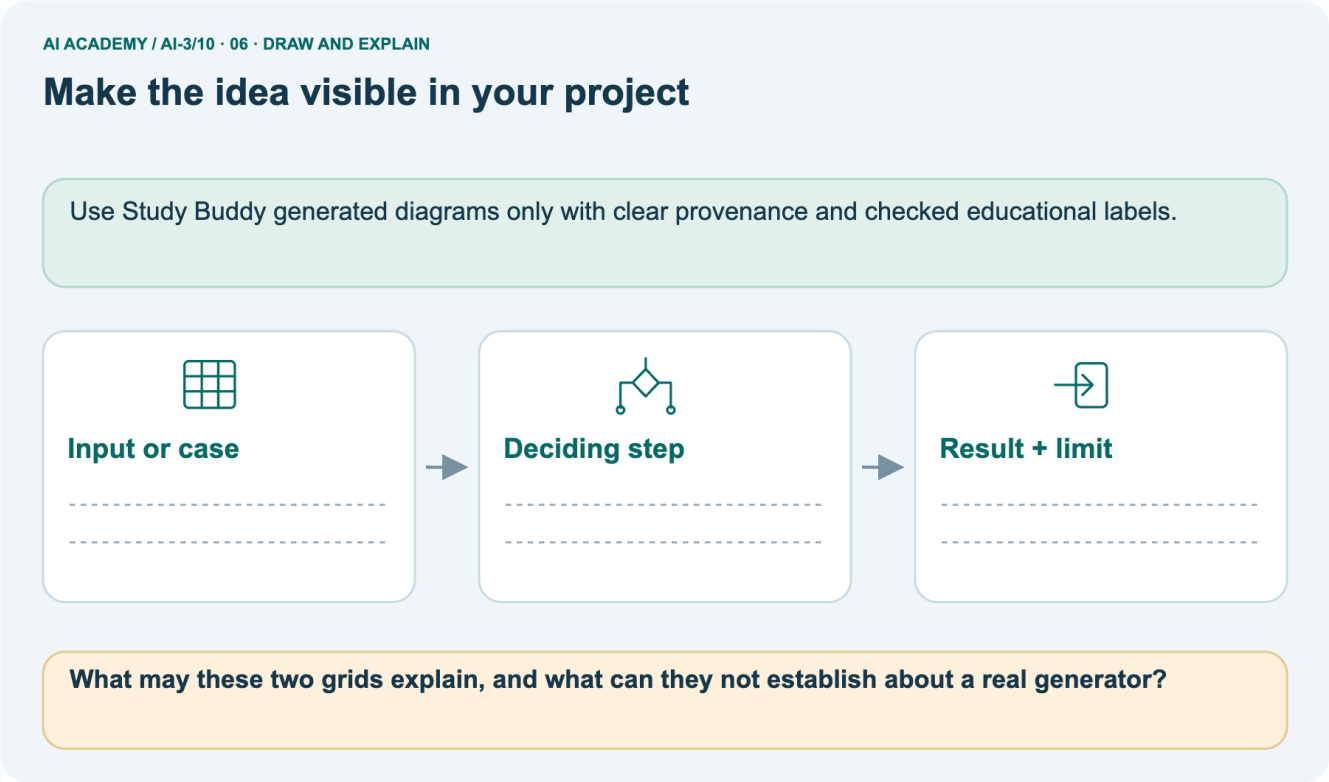
The toy values let us inspect an update without pretending to simulate that trained **process**. An attractive final image still needs checks for labels, requested relations and factual claims.

Limit: Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use Study Buddy generated diagrams only with clear **provenance** and checked educational labels. Draw your **input**, deciding step, result and limit.

Supported question: What may these two grids explain, and what can they not establish about a real generator?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Image Generation Models: the case

02. Follow the deciding steps

03. Illustrative grids, not executed model outputs

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.